CHE 313
HW 5

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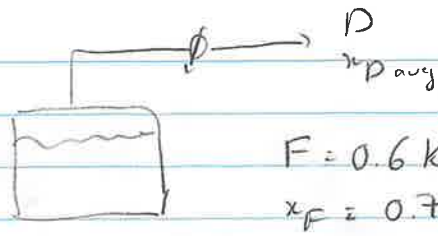
10

9.D.1 (2)

Target comp: methanol



$F = 0.6$ kmol
$x_F = 0.70$
$x_f = 0.10$

Rayleigh eqn:

$$\ln\left(\frac{W_f}{F}\right) = \underbrace{-\int_{x_f}^{x_F} \frac{dx}{y-x}}_{\text{area}}$$

a)

Calculate area using Simpson rule:

$x_f = 0.1 \Rightarrow y_f = 0.418 \Rightarrow \frac{1}{y_f - x_f} = 3.145$

$x_F = 0.7 \Rightarrow y_F = 0.87 \Rightarrow \frac{1}{y_F - x_F} = 5.882$

$x_m = \frac{0.7 + 0.1}{2} = 0.4 \Rightarrow y_m = 0.729 \Rightarrow \frac{1}{y_m - x_m} = 3.040$

$$-\int_{x_f}^{x_F} \frac{dx}{y-x} = \frac{x_F - x_f}{6}\left(f(x_F) + 4f(x_m) + f(x_f)\right)$$

$$= -\frac{0.7 - 0.1}{6}\left(5.882 + 4 \cdot 3.040 + 3.145\right) = -2.119$$

$$\Rightarrow \ln\left(\frac{W_f}{F}\right) = -2.119 \Rightarrow W_f = F e^{-2.119} = 0.6 \times e^{-2.119}$$

$W_f = \boxed{0.0721 \text{ kmol}}$

$F = W_f + D_{tot} \Rightarrow D_{tot} = F - W_f = 0.6 - 0.0721 = \boxed{0.528 \text{ kmol}}$

$F x_F = W_f x_f + D_{tot} x_{D\,avg} \Rightarrow x_{D\,avg} = \frac{F x_F - W_f x_f}{D_{tot}} = \boxed{0.782}$

b) Gaussian quadrature formula

$$\ln\left(\frac{W_f}{F}\right) = -\int_{x_f}^{x_F} \frac{dx}{y-x} = -\frac{h}{9}\left[5f(x_{mid} - h\sqrt{0.6}) + 8f(x_{mid}) + 5f(x_{mid} + h\sqrt{0.6})\right]$$

$$h = \frac{x_F - x_f}{2} = 0.3 \quad x_{mid} = \frac{x_F + x_f}{2} = 0.4$$

$$x_{mid} - h\sqrt{0.6} = 0.4 - 0.3\sqrt{0.6} = 0.168$$

$$x_{mid} + h\sqrt{0.6} = 0.4 + 0.3\sqrt{0.6} = 0.632$$

Interpolate for $y(x = 0.168)$:

$$\frac{0.2 - 0.168}{0.168 - 0.15} = \frac{0.579 - y(x=0.168)}{y(x=0.168) - 0.517} \Rightarrow y(x=0.168) = 0.539$$

$$\Rightarrow \left.\frac{1}{y-x}\right|_{x=0.168} = 2.695$$

Interpolate for $y(x = 0.632)$

$$\frac{0.7 - 0.632}{0.632 - 0.6} = \frac{0.87 - y(x=0.632)}{y(x=0.632) - 0.825} \Rightarrow y(x=0.632) = 0.839$$

$$\left.\frac{1}{y-x}\right|_{x=0.632} = 4.831$$

$$\Rightarrow \ln\left(\frac{W_f}{F}\right) = -\frac{0.3}{9}\left[5 \cdot 2.695 + 8 \times 3.0395 + 5 \times 4.831\right] = -2.065$$

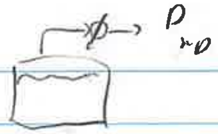
$$\Rightarrow W_f = F\exp(-2.065) = \boxed{0.0761 \text{ kmol}}$$

$$\Rightarrow D_{tot} = F - W_f = 0.6 - 0.0761 = \boxed{0.524 \text{ kmol}}$$

$$x_{D_{avg}} = \frac{F x_F - W_f x_f}{D_{tot}} = \boxed{0.787}$$

9.D.6:

2.6



$x_F = 0.6$

$x_f = 0.3$

target comp: 1,2 dichloroethane

a) $F = 1.3$ kmol

$$\ln\left(\frac{W_f}{F}\right) = \frac{1}{\alpha - 1}\ln\left[\frac{x_f(1-x_F)}{x_F(1-x_f)}\right] + \ln\left(\frac{1-x_F}{1-x_f}\right)$$

$$W_f = F\exp\left(\frac{1}{2.4-1}\ln\left[\frac{0.3(1-0.6)}{0.6(1-0.3)}\right] + \ln\left(\frac{1-0.6}{1-0.3}\right)\right)$$

$W_f = 0.304$ kmol

$D = F - W_f = 1.3 - 0.304 = \boxed{0.996\text{ kmol}}$

$F x_F = W_f x_f + D_{tot}\, x_{D\,avg}$

$$\Rightarrow x_{D\,avg} = \frac{F x_F - W_f x_f}{D_{tot}} = \frac{1.3 \cdot 0.6 - 0.304 \cdot 0.3}{0.996}$$

$\Rightarrow x_{D\,avg} = \boxed{0.692}$ ~~kmol~~

.6914

b) $F = 3.5$ kmol

$\Rightarrow W_f = \boxed{0.817\text{ kmol}}$

$D_{tot} = F - W_f = 3.5 - 0.817 = 2.683$ kmol

$$x_{D\,avg} = \frac{F x_F - W_f x_f}{D_{tot}} = \boxed{0.691}$$ ~~kmol~~

0.6916 ?

c) $F = 2$ kmol

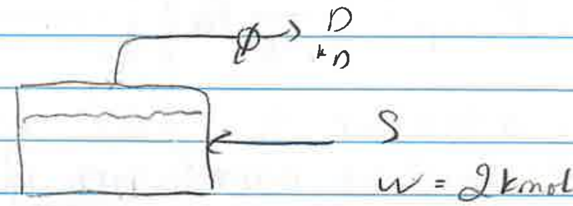
$x_{D\,avg} = 0.75$

$$\begin{cases} W_f = F\exp\left[\frac{1}{\alpha-1}\ln\left(\frac{x_f(1-x_F)}{x_F(1-x_f)}\right) + \ln\left(\frac{1-x_F}{1-x_f}\right)\right] \\ x_{D\,avg} = \dfrac{F x_F - W_f x_f}{F - W_f} \end{cases}$$

$$\Rightarrow \left\{ \begin{array}{l} W_f = 2\exp\left[\frac{1}{2.4-1}\ln\left[\frac{x_f(1-0.6)}{0.6(1-x_f)}\right] + \ln\left(\frac{1-0.6}{1-x_f}\right)\right] \\ 0.75 = \dfrac{2\times 0.6 - W_f x_f}{2 - W_f} \end{array}\right.$$

$$\Rightarrow \left\{ \begin{array}{l} W_f = \boxed{1.211\,\text{kmol}} \\ x_f = \boxed{0.502} \end{array}\right.$$

9.D.11



$$\frac{S}{W} = \int_{x_{w_f}}^{x_{w_i}} \frac{dx_w}{y}$$

$x_{w_i} = 1$

$x_{w_f} = 0.6$

Use VLE data from table 8-5 for this problem

x	y	$\frac{1}{y}$
1.0	1	1
0.9	0.58	1.72
0.8	0.435	2.30
0.7	0.34	2.94
0.6	0.30	3.33

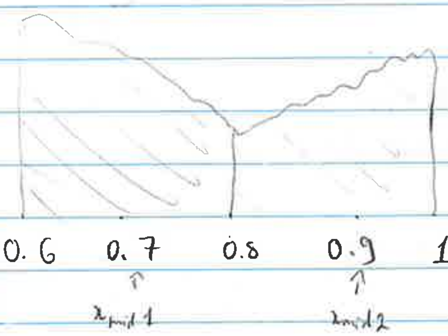
Simpson integration:

$$\int_{0.6}^{1} \frac{dx_w}{y} = \frac{1-0.6}{6}\left(3.33 + 4\times 2.30 + 1\right)$$

$$= 0.902$$

$$\Rightarrow \frac{S}{W} = 0.902 \Rightarrow S = 0.902W = \boxed{1.804\ \text{kmol}}$$

Two step integration



2 step integration

$$\int_{0.6}^{1} \frac{dx}{y} = \int_{0.6}^{0.8} \frac{dx}{y} + \int_{0.8}^{1} \frac{dx}{dy}$$

$$= \frac{0.8-0.6}{6}\left(3.33 + 4\times 2.94 + 2.30\right) + \frac{1-0.8}{6}\left(2.30 + 4\times 1.72 + 1\right)$$

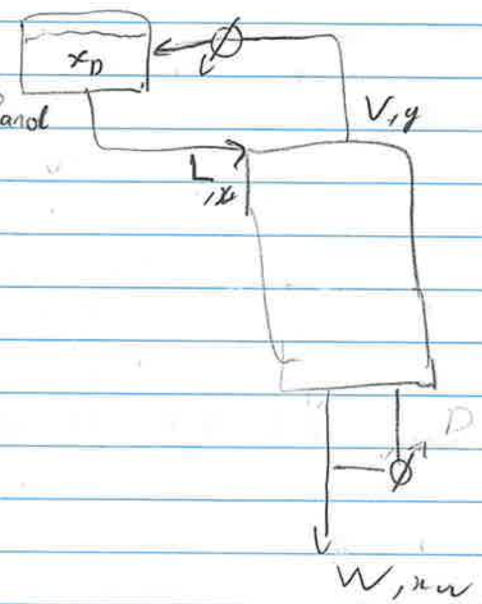
$$= 0.920 \Rightarrow S = 0.920W = \boxed{1.840\ \text{kmol}}$$

There is about 2% difference between 2 methods, can use 1 step integration to be faster

It is reasonable to assume 2step integration is more accurate, $1-\frac{1.748}{1.805} \approx 3.16\%$

=> The difference is about 3%, which might be considered neglectible, we can use one step integration without too much error in calculating added water

9.D.3. D



Target component methanol

$F = 1$ kmol

$x_F = 0.57$

$x_w = 0.15$

$L_0/D = 1.85$

Column M.B:

$$\begin{cases} V = L + D \\ Lx = Vy + W x_D \end{cases}$$

$$\Rightarrow y = \frac{L}{V}x - \frac{W}{L}x_D$$

$$y = \frac{L}{V}x + \frac{V-L}{L}x_D$$

$$y = \frac{L}{V}x + \left(1 - \frac{L}{V}\right)x_w$$

$$\frac{L}{V} = \frac{L}{L+D} = \frac{L/D}{L/D+1} = \frac{1.85}{1.85+1} = 0.65$$

$$\Rightarrow y = 0.65x + 0.35x_D$$

Derived Rayleigh eqn for this configuration:

$$\ln \frac{W_f}{F} = -\int_{x_{wf}}^{x_f} \frac{dx_D}{x_D - x_w}$$

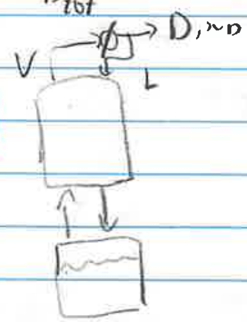
$$= -\int_{0.15}^{0.57} \frac{dx}{y-x} = -\frac{0.57-0.15}{6}\left(3.077 + 4 \cdot 2.096 + 1.754\right) = -0.925$$

$\Rightarrow W_f = F e^{-0.925} = 0.397$ kmol

$\Rightarrow D_{tot} = F - W_f = 1 - 0.397 = 0.603$ kmol

$$x_{D\,avg} = \frac{F x_F - W_f x_f}{D_{tot}} = \frac{1 \times 0.57 - 0.397 \times 0.15}{0.603} = \boxed{0.847}$$

9.D.16 2



Target comp. method

$F = 10$ kmol
$x_F = 0.4$
$x_f = 0.08$
$x_D = 0.84 \rightarrow$ constant, vary L/D

a) Initial reflux ratio correlate to $x_f = 0.4$ and $x_D = 0.84$

Trial & error with slope

$$\frac{L}{V} = \frac{\Delta y}{\Delta x} = \frac{0.84-0.57}{0.84-0} = 0.3214$$

$$\frac{L}{D} = \frac{L}{V-L} = \frac{L/V}{1-L/V} = \frac{0.3214}{1-0.3214} = \boxed{0.4737}$$

b) Final reflux ratio correlate to $x_f = 0.08$ and $x_D = 0.84$

Trial & error to fit 3 stages : $\frac{0.84-0.13}{0.84-0} = 0.8452$

$$\frac{L}{D} = \frac{L/V}{1-L/V} = \frac{0.8452}{1-0.8452} = \boxed{5.4615}$$

D Total = ?

9.D.16

min L/D

max L/D